Assemblies, constraints & file types

GRAPHICS WITH MR KELLY

Higher · CAD & technology · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ assemble components with mate, align and centre axis
- ☐ use orientate, offset and tangent constraints to fully constrain an assembly
- ☐ explain top-down and bottom-up modelling and sub-assemblies
- ☐ explain STEP, IGES, DXF and 3DS file types

I DO — Worked example

Read how a full-mark answer is built

A wheel is fitted onto an axle and pushed against the side of a bracket, with its outer face level with the bracket's outer face.

1 State the three assembly constraints used. (3 marks)

Full-mark answer each ✓ is one mark

Centre axis, to line the wheel hole up with the axle ✓. Mate, to bring the wheel face against the bracket ✓. Align, to make the outer faces level ✓.

Why marks are not awarded

"Line up, push together and make level."

0 marks: the course specification terms must be used.

WE DO — Guided practice

Use the starters and word bank to help you

An engineer produces orthographic drawings directly from a 3D CAD assembly.

2 Describe two advantages of producing the drawings from the 3D model. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

automatically · dimensions · always · accurate · update · straight · changed · drawings

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which constraint lines up a pin with a hole?
☐ Mate ☐ Align ☐ Centre axis ☐ Fillet

b) Two faces level with each other, facing the same way, is...
☐ Align ☐ Mate ☐ Centre axis ☐ Chamfer

c) Which format would you send to a company using different CAD software?
☐ JPG ☐ STEP ☐ DOCX ☐ PNG

d) Modelling parts inside the assembly so changes update related parts is...
☐ Bottom-up ☐ Top-down ☐ Exploded ☐ Suppressed

4 Key terms: write the letter of the matching description next to each term.

1. **Mate**	____	**A** Two faces brought together facing each other.
2. **Align**	____	**B** Two faces made flush, facing the same way.
3. **Centre axis**	____	**C** Lines up the axes of two round features.
4. **Orientate**	____	**D** Parts modelled in the assembly referencing each other; changes update automatically.
5. **Offset constraint**	____	**E** Sets an angle between two parts.
6. **Top-down modelling**	____	**F** Sets a distance between two faces.

A needle cartridge is modelled in position by projecting the hole sizes from the tonearm head (top-down modelling).

5 Explain two benefits of modelling in this way. (2 marks)

A supplier sends a 3D model of a motor to a company that uses different CAD software.

6 State a suitable file format and explain why it is used. (2 marks)

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A tonearm assembly has a base, an axle and a main arm. The arm pivots on the axle and is set at 20° to the base, 13 from the base's inner edge.

7 Describe how the tonearm would be fully assembled. (4 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?